QUALITY INFORMATION DOCUMENT

For
OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream _destabilization_point

Issue: 1.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/11/12	All	Initial version	K.GRASSI	M-I Pujol

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1) INTRODUCTION

a) Product Name

OMI_CIRCULATION_BOUNDARY_ATLANTIC_gulf_stream_destabilization_point

b) Indicator Evaluation Method

The identification of the Gulf Stream path is based on daily Absolute Dynamic Topography (ADT) maps from satellite altimetry averaged to produce monthly maps with a spatial resolution of 1/8°. Here, the Gulf Stream path was identified with the 25 cm SSH contour from detrended ADT according to. e.g., Lillibridge and Mariano (2013), Rossby et al. (2014), Andres (2016). Chi et al. (2021), and Guo et al. (2023).
The yearly Gulf Stream destabilization point is computed following the methodology of Andres (2016). Monthly mean Gulf Stream paths are divided into 0.5° longitude bins, and the yearly latitude variance of the Gulf Stream's position is calculated in each bin, then the destabilization point is defined as the first downstream longitude where the latitude variance reaches 0.42 $(°)^2$, equal to half the maximum variance observed in the aggregate 1993–2022 data set. This threshold marks the transition from a stable, detached jet to an unstable, meandering jet.

For each of the 12 monthly mean paths each year, the destabilization point is identified, producing a distribution of 12 values. From this monthly distribution, the 95% confidence interval of the annual destabilization point is derived. To minimize spurious signals due to high-frequency variability, a 5-year running mean is applied to the yearly destabilization time series for both longitude and latitude.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
SEALEVEL_GLO_PHY_L4_REP_OBSERVATIONS_008_047	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-SL-QUID-008-032-068.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-SL-PUM-008-032-068.pdf
SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046	QUID: https://documentation.marine.copernicus.eu/QUID/CMEMS-SL-QUID-008-032-068.pdf PUM: https://documentation.marine.copernicus.eu/PUM/CMEMS-SL-PUM-008-032-068.pdf

2) SCIENTIFIC VALIDATION

d) Physical Signals

The Gulf Stream Destabilization Point (GSDP) indicator was implemented and validated in Sánchez-Román et al. (2024) as part of the *Copernicus Marine Ocean State Report 8 (OSR8)*. The method identifies the longitude at which the Gulf Stream (GS) transitions from a stable, coherent western boundary current into a meandering, eddy-rich regime. Using sea surface height (SSH) fields for 1993–2021, Sánchez-Román et al. (2024) showed that this transition consistently occurs downstream of Cape Hatteras, where latitude variance sharply increases (Figure 1a). This spatial pattern matches the classical description of the GS evolving into an unstable jet (Hogg, 1992; Bane and Dewar, 1988), with variance maxima in the region of intensified meander growth and eddy shedding. The location of the GSDP is dynamically consistent with changes in potential vorticity gradients and slope–basin interactions documented in earlier studies (Johns et al., 1995; Savidge and Bane, 1999).

Validation is strengthened by the strong agreement between the GSDP and the longitude of maximum low-frequency Eddy Kinetic Energy (EKE), with a correlation coefficient of 0.88 and a mean longitudinal offset of less than 3° (Figure 1b). This confirms that the indicator effectively captures the onset of enhanced mesoscale activity. Furthermore, the SSH-based destabilization point corresponds to a robust surface expression of deeper structural changes in the Gulf Stream: its longitudinal variability closely matches the location of increased variability in the depth of the 12 °C isotherm at 450 m, as observed from hydrographic sections along 38° N (Figure 2). This agreement confirms that surface meandering reflects subsurface changes in thermocline structure and jet coherence.

The GSDP also exhibits sensitivity to basin-scale atmospheric forcing. A statistically significant correlation ($r \approx 0.7$) was found between the annual GSDP position and the North Atlantic Oscillation (NAO) index, indicating that large-scale atmospheric variability modulates the location of the stability–instability transition. This supports the interpretation of the GSDP as a climate-relevant metric for monitoring Gulf Stream dynamics.

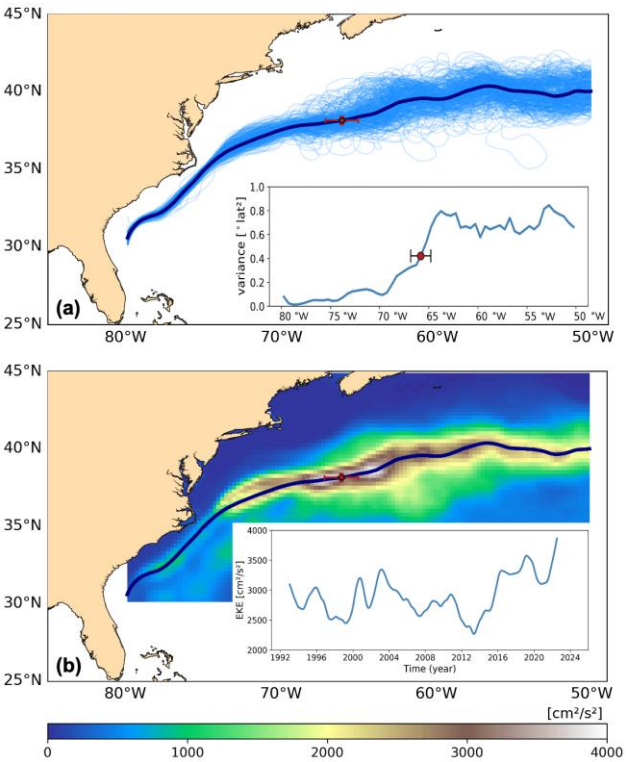

Figure 1: Panel (a) displays the Gulf Stream paths based on the 25 cm SSH contour from altimetry (product ref. no. 1) showing monthly mean (pale blue) and a 1993–2022 overall mean (blue). The inset displays the variance in latitudinal position of the monthly mean Gulf Stream paths (1993–2022) as a function of downstream longitude. Panel (b) displays the 1993–2022 overall mean EKE in the Gulf Stream region and the 1993–2022 overall Gulf Stream mean path (blue). The inset shows the aggregated EKE associated with the Gulf Stream. The red dot in panels (a) and (b) indicates the mean destabilization point (see text for more details). The confidence interval (at 95 % confidence level) of the destabilization point in both longitude and latitude is also displayed.

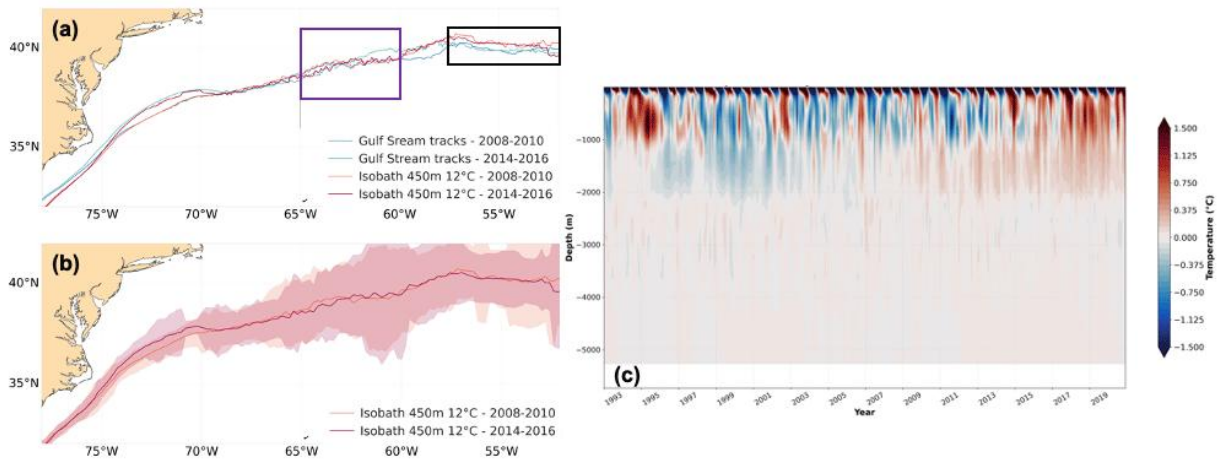

Figure 2: (a) The mean 12 °C isotherm (iso12) for 2008–2010 (orange) and 2014–2016 (red) computed from ocean reanalysis data (product ref. no. 2) superimposed to the SSH derived Gulf Stream pathway computed from altimetry data (product ref. no. 1) for the same periods (blue and green, respectively). (b) The mean and standard deviation (2σ) of monthly pathways estimated using the iso12 at 450 m depth computed from ocean reanalysis data (product ref. no. 2) for the periods 2008–2010 (orange) and 2014–2016 (red). Panel (c) displays the temporal evolution of temperature anomalies in the water

column (°C) in the surroundings (downstream) of the mean Gulf Stream's destabilization point (purple box) computed for the reference period 1993–2020.

e) Methodology Uncertainty

The GSDP is subject to uncertainties stemming from both the observing system and the detection methodology. The 1/8° nominal resolution from SSH fields used for path detection limits the ability to resolve submesoscale variability and sharp frontal gradients, and the objective mapping applied to produce gridded fields may introduce spatial smoothing that is particularly relevant in strong-gradient regions such as western boundary currents. Such smoothing can slightly shift the detected path or attenuate the latitude variance, potentially biasing the estimated destabilization point. The variance threshold criterion (0.42 $(°)^2$), set as half of the aggregated maximum variance following Andres (2016) method, is applied uniformly across years to ensure temporal consistency; however, this assumes a stationary eddy kinetic energy (EKE) regime. Documented increases in Gulf Stream EKE (Barceló-Llull et al., 2025) could elevate background variance levels, leading to an earlier downstream crossing of the threshold without necessarily reflecting a real change in physical destabilization. Finally, the use of monthly mean paths in the variance computation reduces high-frequency noise but may smooth out short-term meander or ring-detachment events, while the confidence interval estimation, based on 30 daily paths per month, accounts for intramonthly variability but may underestimate low-frequency or systematic errors associated with long-term mapping changes.

Despite these limitations, the robustness of the GSDP is enhanced by its physical grounding (as a proxy for jet stability), its temporal coverage over the full altimetric period, and its sensitivity to large-scale atmospheric forcing. However, continued monitoring of the performance of the underlying altimetric products and exploration of adaptive or dynamic thresholds may be warranted as the ocean's mesoscale energy distribution evolves under climate change.

For further detail on the methodology, validation, and limitations, readers are referred to Sánchez-Román et al. (2024), the OSR8, and the references therein.

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